

Jiawei Zhang | motto: altiora peto

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Education

Academic Qualifications

- **University of Liverpool** **Liverpool, UK**
M.Sc. Computer Science, with distinction honor *2014.09–2015.12*
- **Sichuan University** **Chengdu, China**
B.Sc. Applied Mathematics *2010.09–2014.07*

Student Projects

- **Master's Project:** *Semantic Relation Extraction Using Long Short-Term Memory (LSTM)*
- **Bachelor's Project:** *Discrete Particle Swarm Optimization in Resources Scheduling of Public Services*

Industrial Experience

- **Neurogin Inc (neurogin.com)** **Chengdu, China**
Senior Research Engineer / Lead of Data&Algorithms Team *2020.11 – now*
- **Nuance/Cerence (nuance.com)** **Chengdu, China**
Deep Learning Research Engineer *2018.07 – 2020.11*
- **Weiheng Technology Inc (ynmaker.com)** **Kunming, China**
Data Mining Engineer *2017.08 – 2018.07*
- **Shu Rui Tech, part of MDQ (meddataquest.com)** **Hangzhou, China**
NLP Research Engineer *2016.02 – 2017.06*

Projects

- **EEG Acquisition & Analysis Platform** lead of data & algorithms
An integrated system for collecting and analysing high-quality human Electroencephalography, which is real-time pain level quantification & prediction in clinical procedures.
- **ASR-oriented Deep Learning Toolkit** maintainer of AM training
A suite of tools designed specifically for building DL models for speech recognition based on C++ and Python.
- **Medical NLP Engine for EHR Analysis** major developer
A self-developed NLP engine that achieves stable and highly accurate (85-90%) recognition (NER) for different medical texts in EHR, which combines computational linguistics, knowledge inference, machine learning and deep learning techniques.
- **Medical Knowledge Graph for Entity Encoding** major developer
A medical entity encoding & inference system that has inference tools that resemble human reasoning. The system is designed with logic and statistics to draw inferences that conform to all knowledge rules in the medical knowledge graph.
- **Invasive Plants Detection** major developer
An automatic invasive plants recognition system that is delivered with a DenseNets model, which achieves classification accuracy around 91% on the test set. The work is in collaboration with the Academy of Agricultural Science (Yunnan).
- **The 'JeeFarm' Farming Bot** project owner
A programmable and automated farming robot (optimized) based on the FarmBot project, it is capable for planting small-sized vegetables and strawberries, within which the insect pests can be recognized by a needle camera.

Technical & Personal Skills

- **Development:** Experienced in Python, Java, Shell. Basic in C++, Groovy, Matlab. Familiar with PyTorch, TensorFlow, DeepLearning4j, Apache UIMA, Docker, Git, etc.
- **Certificates & Learning:** Deep Learning Specification, DeepLearning.ai TensorFlow Developer (Coursera)
- **Open Source Effort:** Minor contributor of "MNE-python", member of "createamind" (Github).
- **Language:** Chinese: native speaker; English: fluent in speaking & writing

Interests & Activities

- **Contests:** The Space App Challenge 2015, NASA, United Kingdom; The 2016 CEGS N-GRID NLP Challenge for Clinical Data, Harvard Medical School, United States.
- **Research Interests:** Natural Language Processing, Neural Networks, Time-series Analysis, Generative Learning and Multi-modal Representation Learning.
- **Daily Hobbies:** Badminton, Cycling, Hiking & Camping, Lure Fishing.

教育经历

- 利物浦大学
计算机科学硕士, 杰出学位
主要课程: 数据库系统, 网络编程, 面向对象软件开发, 算法导论, 数据挖掘, 计算智能, 神经网络
利物浦, 英国
2014.09-2015.12
- 四川大学
应用数学学士,
主要课程: 数学分析, 高等代数, 概率论, 统计学, 随机过程, 实变函数, 复变函数, 一般拓扑学
成都, 中国
2010.09-2014.07

学生项目

- 硕士毕业项目: 基于深度学习的语义关系抽取
- 学士毕业项目: 基于离散粒子群算法的公益资源调度优化

工作经历

- 新源生物电子科技有限公司
资深研发工程师/数据与算法组负责人
成都, 中国
2020.11 - 至今
- 纽昂思软件技术有限公司(Nuance/Cerence)
深度学习研发工程师
成都, 中国
2018.07 - 2020.11
- 唯恒基业科技有限公司
数据挖掘工程师
昆明, 中国
2017.08 - 2018.07
- 数睿科技有限公司(Med Data Quest)
自然语言处理研发工程师
杭州, 中国
2016.02 - 2017.06

项目经历

- 人体多通道脑电采集及分析系统
一个收集和分析高质量的人类脑电图的综合系统, 用于临床手术过程中实时的疼痛等级量化及预测。
lead of data & algorithms
- 电子病历NLP分析引擎
一个自主开发的医疗NLP引擎, 对EHR中的不同医疗文本实现了稳定和高度准确(85-90%)的识别(NER), NLP任务包括: 医学实体识别、语义关系抽取、歧义消除、病历搜索、疾病分类、病人相似度分析等, 实现中结合了计算语言学、知识推理、机器学习和深度学习技术。
major developer
- ASR深度学习训练工具
一个面向语音识别的深度学习工具箱。在Nuance/Cerence期间, 负责自研语音识别算法及深度学习平台的研发及维护工作, 成果在公司车载智能语音助手、海外医疗智能语音助手等核心的业务模块得到广泛应用。
maintainer of AM training
- 入侵性植物检测系统
一个自动入侵植物识别系统, 该系统采用DenseNets模型, 在测试集上实现了约91%的分类精度, 该工作是与云南农业科学院合作进行的。
major developer
- 智能音乐播放器语音助手
一个智能音乐播放器Flow内部语音助手EVA, 实习期间参与其语义理解引擎开发, 包括用户意图识别及分类、关键词抽取、对话系统等功能研发。EVA是渡鸦科技智能硬件产品Raven Home的核心交互模块, 在被百度收购之后, 融入旗下人工智能产品小度智能语音助手产品之中。
NLP engineer intern

技术与个人技能

- 编程开发: 熟练: Python, Java, Shell; 基础: C++, Groovy, Matlab; 熟悉: PyTorch, TensorFlow, Deeplearning4j, Apache UIMA, Docker, Git等
- 通用技能: 良好的演示、沟通技能, 有良好的合作意识, 态度积极认真。
- 语言能力: 中文: 母语; 英语: 流利, 具备一定学术写作能力。

兴趣与活动

- 会议及比赛: The Student Enterprise Conference 2015, NACUE, United Kingdom; The Space App Challenge 2015, NASA, United Kingdom; The 2016 CEGS N-GRID NLP Challenge for Clinical Data, Harvard Medical School, United States.
- 研究兴趣: 自然语言处理、神经网络、生成式模型、时间序列数据分析、深层强化学习
- 日常兴趣: 羽毛球、骑行、徒步、露营、路亚